



Bhartendu Mishra, Managing Director, Arrow Electronics, India

2019, along with programmes such as Make in India, Digital India, and Future Skills, emphasises workforce development, digital infrastructure, and indigenous innovation. Some recent specialised support includes the MeitY-Meta XR Accelerator, quantum computing labs with Amazon Web Services (AWS), and partnerships like the IIT Indore-CIMEI MoU.

Further, the Advanced Chemistry Cell (ACC) PLI (₹181 billion) supports battery manufacturing, while electronics research and development (R&D) is backed by a dedicated ₹320 million research fund.

Moreover, India's ESDM sector has attracted ₹490.68 billion (US\$5.67 billion) in FDI (2000-2024). Three years ago, a *Frost & Sullivan* report predicted that the ESDM market in the country would grow from ₹2.65 trillion (approximately US\$36 billion) in FY21 to ₹9.96 trillion (approximately US\$135 billion) by FY26. The share of Indian companies is expected to jump from ₹1.7 trillion (approximately US\$14 billion) to ₹5.97 trillion (approximately US\$81 billion), driven by

design-led manufacturing, import substitution, and increasing demand for localised solutions.

As per a December 2024 report by *The Economic Times*, India's ESDM industry, valued at ₹2.09 trillion in FY23, is projected to grow at a 34% compound annual growth rate (CAGR) to reach ₹9.09 trillion by FY28. A recalibration, not a slowdown, is needed, as earlier projections overshot near-term growth. But what is the contribution to this growth from the

Indian perspective?

It must be noted that research and design are two different games altogether; the funding support in India is primarily allocated to research, and this support is typically given to academic institutions. And, somehow, it has never resulted in any product/system design and reached the Indian industries.

There is no comprehensive/structured funding scheme to promote product and systems designs that could be mass-manufactured by the Indian industrial ecosystem, be it for local consumption or the global market.



Jaswinder Ahuja, ex-Corporate Vice President of Cadence Design Systems (India)

The constraints inhibiting innovation and design in India

With a population of 1.4 billion and a massive consumer base, the country presents a natural market for electronics. However, this potential remains largely untapped. "We have the market, we have the talent, so why cannot we do design, making all in India?" pointed out T.S. Shankar, currently the Sales Director at Analog Devices.

Government incentives like PLI and DLI exist, but struggle with execution and limited uptake. The design gap is glaring: while Indian engineers play a crucial role in global tech development, including quantum computing, domestic ownership of innovation remains minimal.

"You can manufacture 100,000 iPhones here, but the question is what is your contribution from the Indian perspective?" noted Shankar. Capturing the opportunity of a \$500 billion electronics and services market by 2040 requires urgent alignment between policy, talent, and execution. "There is a window of opportunity. The question is if we can do that, and how?" he said.

This urgency was echoed by another industry veteran, Bhartendu Mishra, Managing Director-India at Arrow Electronics, who highlighted India's fast-growing semiconductor consumption, at \$41 billion annually, or 8% of global demand, as a signal

that the country can no longer rely on being a consumption-led market. "What happened in the first 20 years doubled in the next eight and doubled again in the next five," he said.

But growth without innovation is unsustainable. Despite producing \$100 billion worth of electronics and exporting